NETWORKS, RANDOM WALKS AND ITERATION ON ONE PAGE

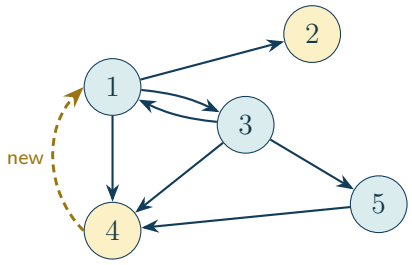
PageRank: let the links vote

Share a score, collect the incoming shares, repeat. One new link can change the leader.

1 / READ THE NETWORK

A link passes on a share of its source's score

Each page splits its current score equally among its outgoing links. Page 1 sends a third to each of $2, 3, 4$; page 3 sends a third to each of $1, 4, 5$; page 5 sends all of its share to 4.



Solid arrows: original network.
Gold dashed arrow: add $4 \to 1$ later.

$$H = \begin{pmatrix} 0 & 0 & \frac{1}{3} & 0 & 0 \\ \frac{1}{3} & 0 & 0 & 0 & 0 \\ \frac{1}{3} & 0 & 0 & 0 & 0 \\ \frac{1}{3} & 0 & \frac{1}{3} & 0 & 1 \\ 0 & 0 & \frac{1}{3} & 0 & 0 \end{pmatrix} \qquad \begin{matrix} H_{ij}: \\ \text{share sent} \\ \text{from } j \text{ to } i. \end{matrix}$$

Columns send; rows receive. For instance, $H_{4,5} = 1$ because $5 \to 4$ is page 5's only link. Columns 2 and 4 are zero: these pages are **dead ends**.

2 / KEEP ALL THE SCORE IN CIRCULATION

Repair dead ends, then allow jumps

At a dead end, jump uniformly to any of the five pages, including the current one. Replace each zero column of H by

$$\mathbf{u} = \tfrac{1}{5}(1, 1, 1, 1, 1)^{\mathsf{T}}.$$

Call the repaired matrix S. Every column now sums to 1: multiplying by S preserves the total score.
Choose a damping factor $\alpha = 0.85$. With probability 0.85 follow the rule in S; with probability 0.15 jump uniformly:

$$\boxed{\mathbf{p}^{(n+1)} = 0.85S\mathbf{p}^{(n)} + 0.15\mathbf{u}.}$$

Here $p_i^{(n)}$ is page i's score after n updates. The five scores always sum to 1.

3 / DO ONE UPDATE BY HAND

Start everyone equally at 0.2

Use $\mathbf{p}^{(0)} = \mathbf{u}$. Page 4 collects shares from $1, 3, 5$. Dead ends $2, 4$ share their combined 0.4 equally across all pages. Each page also receives $0.15/5 = 0.03$ from random jumps:

$$\begin{aligned} p_4^{(1)} &= 0.03 + 0.85\left(\frac{0.2}{3} + \frac{0.2}{3} + 0.2 + \frac{0.4}{5}\right) \\ &= 0.381333\ldots \end{aligned}$$

The other four scores are $0.154666\ldots$.
Update simultaneously: calculate every new score from the same old vector. Then replace the old vector and repeat, keeping full precision.

4 / REPEAT ON THE ORIGINAL NETWORK

The scores settle; page 4 leads

n	p_1	p_2	p_3	p_4	p_5
0	.2000	.2000	.2000	.2000	.2000
1	.1547	.1547	.1547	.3813	.1547
2	.1649	.1649	.1649	.3402	.1649
3	.1626	.1626	.1626	.3495	.1626
5	.1630	.1630	.1630	.3479	.1630
10	.1630	.1630	.1630	.3478	.1630
∞	.1630	.1630	.1630	.3478	.1630

The limiting vector is exactly

$$\mathbf{p}^* = \tfrac{1}{92}(15, 15, 15, 32, 15)^{\mathsf{T}}.$$

Thus 4 ranks first; $1, 2, 3, 5$ tie. Its 34.78% share is the random surfer's long-run fraction of time on page 4.

5 / ADD THE GOLD LINK AND START AGAIN

One new arrow: $4 \to 1$

Page 4 now sends all its score to 1. In S, change column 4 from $\mathbf{u}$ to $(1, 0, 0, 0, 0)^{\mathsf{T}}$. Page 2 remains a dead end. Restart at $\mathbf{p}^{(0)} = \mathbf{u}$; keep $\alpha = 0.85$.

n	p_1	p_2	p_3	p_4	p_5
0	.2000	.2000	.2000	.2000	.2000
1	.2907	.1207	.1207	.3473	.1207
2	.3799	.1329	.1329	.2696	.0847
3	.3194	.1602	.1602	.2699	.0902
5	.3263	.1492	.1492	.2783	.0970
10	.3309	.1488	.1488	.2738	.0976
∞	.3305	.1490	.1490	.2741	.0975

Now $1 > 4 > 2 = 3 > 5$ in score. Although 4 still has three incoming links, the new feedback route makes 1 the leader.

6 / THE LINK TO MATRICES AND FIXED-POINT ITERATION

A whole-network judgement from a repeated local rule

Write $G = 0.85S + 0.15\mathbf{u}\mathbf{1}^{\mathsf{T}}$, where $\mathbf{1} = (1, 1, 1, 1, 1)^{\mathsf{T}}$. Since $\mathbf{1}^{\mathsf{T}}\mathbf{p} = 1$, the update is $\mathbf{p}^{(n+1)} = G\mathbf{p}^{(n)}$. At the limit, $\mathbf{p}^* = G\mathbf{p}^*$: PageRank is a normalised **eigenvector with eigenvalue** 1. The uniform jumps ensure a unique limit from any starting probability vector.
To stop numerically, test $\sum_i |p_i^{(n+1)} - p_i^{(n)}| < 10^{-8}$. The scores measure attention in this model; their ordering gives the ranking.

Model. Five pages, equal sharing along outgoing links, uniform dead-end redistribution and jumps, $\alpha = 0.85$. Tables are rounded to four decimal places.
Network. Redrawn from the supplied Math Mindset illustration.
Further reading. Stanford CS246, *PageRank*; Google Cloud, *PageRank*: random walks and iteration.

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